

Certificate of Participation

Riya Dakshesh Modi

has successfully participated in the OPEN vhb online course

Industry 4.0: Aspects of Technology Advanced, Part 1 (InnoPeerAVM Advanced Training)

The online self-tests required to issue this certificate have been passed.

Learning Content:

1. Additive Manufacturing

- Basic principles of additive manufacturing (AM) technologies and process chains
- Economic aspects of AM and new business models
- Potentials of AM within the topic of Advanced Manufacturing

2. Big Data

- Data as important asset in the context of Advanced Manufacturing
- Big Data and Artificial Intelligence: Two closely connected concepts
- Barriers vs. Potentials: Basic methods and successful pilot projects

3. Connectivity

- Basics of network technology for the Industrial Internet of Things (IIoT)
- Differences between current and future connectivity concepts on the shop floor
- Basics of OPC UA as possible connectivity technology for the IIoT

4. Security

- Plant floor cyber security: Threats in production environments
- Data security as strategic success factor for Advanced Manufacturing
- Current standards and future development in industrial cyber security

Certificate of Participation

Riya Dakshesh Modi

has successfully participated in the OPEN vhb online course

Industry 4.0: Aspects of Technology Advanced, Part 2 (InnoPeerAVM Advanced Training)

The online self-tests required to issue this certificate have been passed.

Learning Content:

1. Additive Manufacturing (AM)
 - History and today's market in the context of development of prominent AM technologies
 - Technologies overview
 - Implementation paths of AM technologies in modern companies
 - Standardization in AM technologies
 - Aerial drone photogrammetry
2. Collaborative Robots Introduction
 - Overview of collaborative technologies on the example of the history of collaborative robots
 - Difference between collaborative and industrial robots and when to use them
 - General understanding of collaborative robotics
3. Collaborative Robots
 - ABB and Universal Robots principles and use-cases
 - Practical implementation examples of Universal Robots and ABB collaborative robots
 - Basics of programming of collaborative robots on the example of UR cobots in specific applications
4. Industry 4.0
 - Intelligent sensors and retrofit of existing machines
 - Basic principles of machine learning in industrial applications
 - Condition monitoring and predictive maintenance
 - Simulation, virtual commissioning and digital twins